

Technical & Functional Specifications

Empowered by Nature. Enhanced by Ayurveda.

A comprehensive document covering the architecture, hardware & software stack, feature set, payment-gateway procedure, cart and checkout flow, admin console, data model, security posture, and deployment topology of the Ayutree e-commerce platform.

Project	Ayutree E-commerce Platform
Version	1.0 (MVP)
Author	Ayutree Engineering
Domain	ayutree.com
Markets	India (■ INR)
Document Type	Technical & Functional Specification

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1. Brand & Vision

Ayutree is a premium ayurvedic skin & hair-care house from India, formulating 1000+ year-old recipes into small-batch, hand-packed modern rituals. The brand stands at the intersection of nature and ayurveda — botanical-first, dermatologist-tested, free of parabens, sulphates and shortcuts.

Brand Identity

Name	Ayutree (ayu tree)
Tagline	Empowered by Nature, Enhanced by Ayurveda
Logo Mark	A stylised gold tree with abundant leaves & fruit
Primary Colour	#1A3B32 (deep teal-green)
Accent Colour	#C2A878 (warm muted gold)
Surface	#F9F6F0 (sand background) / #FFFFFF
Typography	Cormorant Garamond (display) + Manrope (body)
Social	Instagram · Facebook · YouTube (@Ayutreecosmetics)

Target Audience

- Indian D2C consumers aged 22–55 with a preference for natural & ayurvedic personal care.
- Gifting customers seeking premium, heritage-inspired beauty.
- Parents shopping for safe, gentle products for children & babies.
- Men seeking grooming exclusives (beard oil, charcoal range).

2. Hardware Specifications

Ayutree is delivered as a cloud-native web application; there is no on-premises hardware for end-users beyond the standard device/browser. Below is the recommended infrastructure and minimum device requirements.

2.1 Server / Cloud Infrastructure

Runtime	Kubernetes pod (Linux x86_64) — auto-scaled
Backend CPU	Min 2 vCPU · Recommended 4 vCPU per replica
Backend RAM	Min 2 GB · Recommended 4 GB per replica
Frontend CPU/RAM	Min 1 vCPU / 1 GB (static SPA via CDN-fronted node)
Database	MongoDB 6.x — Min 4 GB RAM, 50 GB SSD (replica set in prod)
Object Storage	S3-compatible bucket (product images) · TLS · versioning enabled
CDN	Global edge cache (HTTPS, HTTP/3, Brotli, image optimisation)
Load Balancer	L7 (Kubernetes Ingress) with TLS-1.2+ termination
Outbound	Razorpay API · Emergent Auth · Email provider

2.2 End-User Device Requirements

Desktop / Laptop	Any modern OS (Win 10+, macOS 11+, Linux) · 4 GB RAM · 1366×768 or higher
Mobile	Android 9+ or iOS 14+ · 2 GB RAM · 360×640 or higher
Browser	Latest Chrome / Edge / Safari / Firefox (ES2020+ support, cookies enabled)
Internet	Min 1 Mbps for browsing · 5 Mbps for video/imagery
Optional	Webcam & mic not required · Camera for photo upload (mobile)

2.3 Admin / Operations Workstation

Device	Standard laptop / desktop with modern browser
Network	Stable 5 Mbps+ broadband

Recommended Display	13" or larger for the admin dashboard
Optional Printer	For invoices & packing slips (A4/A5 thermal)

3. Software Specifications & Stack

3.1 Frontend Stack

Framework	React 19 · React Router v7 · CRA + craco
Language	JavaScript (ES2022, JSX)
Styling	Tailwind CSS 3.4 · shadcn/ui (radix primitives) · Cormorant + Manrope (Google Fonts)
Animation	Framer Motion 11
Icons	@phosphor-icons/react (light/thin weight)
Data Layer	Axios (with credentials) · TanStack Query
Charts	Recharts 3
Toasts	Sonner
State	React Context (Auth, Cart) + component-level state
Build	yarn · craco start / build

3.2 Backend Stack

Framework	FastAPI 0.110 · Uvicorn (ASGI)
Language	Python 3.11
Database Driver	Motor 3.3 (async MongoDB)
Authentication	PyJWT (HS256) · bcrypt · httpOnly cookies + Bearer fallback
Payments	razorpay (Python SDK)
Validation	Pydantic v2
HTTP Client	requests (for Emergent Auth)
Process Mgr	supervisord

3.3 Third-Party Integrations

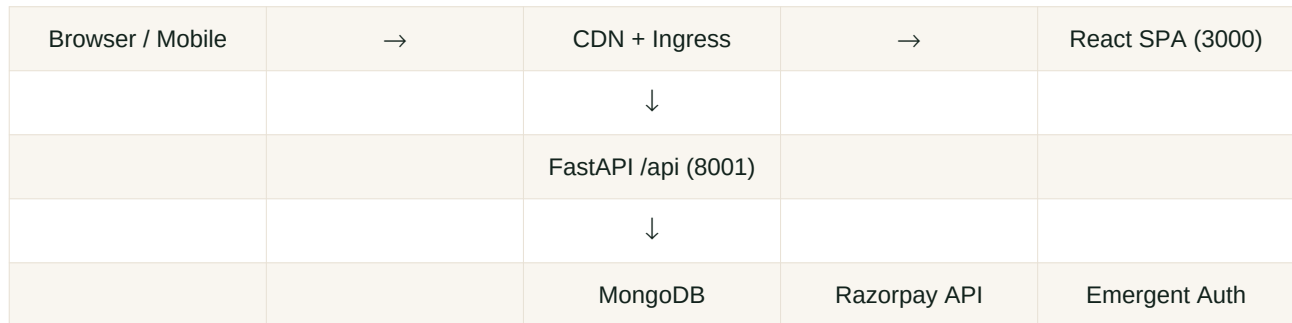
Payments	Razorpay (UPI / Card / NetBanking / Wallets) — INR
Google OAuth	Emergent-managed Google Social Login

Email	Resend (transactional) — planned
Object Storage	S3-compatible — planned for product image upload
Analytics	Built-in dashboard (Recharts) · GA-4 ready (planned)

4. System Architecture

Ayutree follows a classical three-tier SPA architecture. The React single-page app talks exclusively to FastAPI through a stable **/api** contract. All backend services route through Kubernetes ingress, which mounts *port 8001* for */api/** and *port 3000* for the SPA. MongoDB is the single source of truth.

4.1 High-Level Diagram



4.2 Request Lifecycle

- Browser loads SPA from CDN; the SPA reads **REACT_APP_BACKEND_URL** at build time.
- All XHR calls hit **/api/***. Ingress strips/forwards to the FastAPI service.
- FastAPI authenticates the request via httpOnly cookie OR **Authorization: Bearer <jwt>**.
- Resource endpoints query MongoDB via Motor (async) and stream JSON back.
- Payment endpoints additionally call Razorpay's REST API server-side; verification uses HMAC-SHA256 of (order_id|payment_id) with the merchant secret.

5. Application Modules & Features

Storefront (Home)	Hero, brand marquee, asymmetric category bento, featured products, story panel.
Catalogue	10 categories, search, sort (featured / price / name), grid & mobile-first layout.
Product Detail (PDP)	Image, price/MRP, benefits chips, qty selector, Add-to-Bag / Buy-Now, tabs (Ingredients · How-to-Use · Description), customer reviews.
Cart	Persisted per user. Real-time qty update, remove, subtotal/shipping/tax/total breakdown.
Wishlist	Toggle from product card or PDP. Persisted server-side.
Checkout	Address form, payment method (COD / Razorpay), order summary, secure submit.
User Account	Order history, status & payment badges, item thumbnails, profile basics.
Authentication	Email/password JWT + Emergent Google social login.
Admin Dashboard	Revenue, orders, customers, products tiles · 7-day sales line chart · top-5 products bar chart · low-stock list.
Admin · Products	Full CRUD: create, edit drawer, delete, search, image URL, stock, MRP, featured flag.
Admin · Orders	Table with status dropdown, detail drawer (address & line items), payment/cod indicator.
Admin · Customers	List with order count & lifetime spend per customer.
Reviews (planned)	POST /reviews exists; UI form on PDP planned.
Forgot Password (planned)	Email link with one-hour token.
Transactional Email (planned)	Order placed / shipped / delivered via Resend.

6. Product Catalogue (40+ SKUs)

All 41 SKUs from the founder's list have been seeded into MongoDB and grouped into 10 categories. Default MRP/price values are editable from Admin → Products.

Face Care	Charcoal Face Wash · Anti-Acne Face Wash · Anti-Acne Face Cream · Kachur & Jojoba Oil Cream · Skin Whitening Cream · Face Powder · Skin Glow Serum · Milk Drop Serum · Goat Milk & Kumkumadi Oil 15mL
Hair Care	Hair Oil · Ancient Method Hair Oil · No More Tears Shampoo · Hibiscus Shampoo · Coconut Shampoo · Hair Growth Serum · Frizz Free Serum · Her Shine Hair Pack · Hair Color Dye · Deep Conditioning Hair Mask
Body Care	Body Lotion (Bay Leaf) · Talcum Powder · Bath Powder · Under Arm Roll-On
Soaps	Tulasi · ABC · Tomato · Kuppaimeni · Papaya · Neem · Charcoal
Lip Care	Pinky Lip Balm · Yellow Fellow Lip Balm · Purple Lip Balm · Lipstick
Eye Care	Eyelash & Eyebrow Growth Oil · Mascara · Kajal
Men Exclusive	Beard Oil
Kids	Baby Massage Oil · Body Lotion for Kids
Foot Care	Foot Crack Cream
Pain Relief	Pain Relief Oil

Per-Product Fields Stored

- **product_id** · UUID stable identifier
- **name, category, sku** · cataloguing
- **price, mrp** · INR base price & recommended retail
- **stock** · live inventory count (low-stock alert below 20)
- **image** · CDN URL (S3 upload planned)
- **short_description, description, ingredients** · merchandising copy
- **benefits** · array of badge labels (e.g., 'Removes impurities')
- **featured** · boolean flag for homepage
- **rating, reviews[]** · social proof
- **created_at** · ISO 8601 timestamp (UTC)

7. Customer Journey & Cart View

7.1 Full Shopping Journey

- **Discovery** — Browse Home (hero + categories) or land via search.
- **Catalogue** — Filter by category & price; sort by featured/price/name.
- **Product Detail** — Read benefits, ingredients & how-to-use; pick quantity.
- **Add-to-Bag** — Persists item to server-side cart bound to **user_id**.
- **Sign-in (if needed)** — Email/password or Google. Cart is reloaded.
- **Cart Review** — Inspect items, change qty, remove, see live totals.
- **Checkout** — Address form, choose payment (COD / Razorpay), confirm.
- **Payment** — Razorpay modal (UPI/Card/NetBanking) OR COD instant placement.
- **Order Placed** — Cart cleared, success page shows order id, history updated.

7.2 Cart View — Field-by-Field

Item Thumbnail	Square image rendered with placeholder fallback when missing.
Product Name	Click to return to PDP.
Category Tag	Small uppercase ribbon (e.g., FACE CARE).
Unit Price	Formatted as ■ INR with no decimals.
Quantity Stepper	– / number / + buttons; qty ≤ 0 removes the line.
Remove Action	'Remove' button under the stepper.
Line Total	qty × unit price, right-aligned.
Subtotal	Sum of all line totals.
Shipping	Free above ■499, else ■49 flat.
Tax	5% GST on subtotal (illustrative; configurable).
Total	Subtotal + Shipping + Tax.
CTA	'Proceed to Checkout' (primary teal button).

7.3 Cart APIs

GET /api/cart	Fetch the signed-in customer's cart with enriched product details.
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POST /api/cart/add	Add { product_id, qty } — merges with existing line if same SKU.
POST /api/cart/update	Set absolute qty; qty ≤ 0 removes the line.
DELETE /api/cart/{product_id}	Remove a line item.

8. Payment Gateway Procedure (Razorpay)

Ayutree integrates Razorpay — India's leading payment gateway — for instant checkout across UPI, debit/credit cards, net-banking, and popular wallets. All payments are denominated in INR (paise on the wire). For shoppers who prefer it, **Cash on Delivery (COD)** is also supported.

8.1 Razorpay Setup (Merchant)

- Create an account at **razorpay.com** and complete KYC (PAN, GST, bank account, signatory ID).
- Open **Dashboard** → **Settings** → **API Keys** → **Generate** to issue *Key ID* and *Key Secret*.
- Create a **Webhook** for events *payment.captured*, *payment.failed*, *order.paid*; copy the *Webhook Secret*.
- Paste all three values into **/app/backend/.env** and restart the backend service.

RAZORPAY_KEY_ID	rzp_live_XXXXXXXXXXXX (or rzp_test_... for sandbox)
RAZORPAY_KEY_SECRET	32-char merchant secret
RAZORPAY_WEBHOOK_SECRET	configured per webhook endpoint

8.2 End-to-End Payment Procedure

1. Place Order Request

Frontend → POST /api/checkout { address, payment_method:'razorpay' }.

2. Backend Validation

Server verifies session, computes subtotal/shipping/tax/total, persists a 'pending' order with status 'awaiting_payment'.

3. Razorpay Order Creation

Server calls Razorpay **orders.create** with amount × 100 (paise), currency 'INR', receipt=ayutree_order_id. Razorpay returns an *rzp order id*.

4. Open Checkout

Frontend dynamically loads checkout.razorpay.com script and instantiates **new Razorpay(options)** with key_id, order_id, prefill (name/email/phone), theme.

5. Customer Pays

Razorpay modal accepts UPI / Card / NetBanking / Wallet. Razorpay handles 3-D Secure, OTP, OTC.

6. Handler Callback

On success, Razorpay invokes the handler with { razorpay_payment_id, razorpay_order_id, razorpay_signature }.

7. Signature Verification

Frontend → POST /api/payments/verify with those three values. Server recomputes HMAC-SHA256(order_id|payment_id, key_secret) and compares to received signature. Mismatch ⇒ 400.

8. Mark Paid

On match, order.payment_status='paid', status='placed', razorpay_payment_id stored; cart cleared.

9. Webhook Reconciliation

Razorpay simultaneously fires a server-to-server webhook so the order state is consistent even if the browser closes prematurely.

10. Customer Confirmation

User is redirected to /profile?order=AYUyymmddXXXXXX with success badge. Email receipt (planned).

8.3 Refunds & Failure Handling

- **Payment failed** — order remains in 'awaiting_payment'. Customer can retry from /profile.
- **Customer cancels modal** — no charge, no state change.
- **Verification mismatch** — order is marked 'failed'; ops team can investigate.
- **Refund** — initiated from Razorpay dashboard or POST /payments/{id}/refund (planned); reflected in admin order detail.
- **COD** — order placed immediately as 'placed'; payment_status='pending' until delivery cash collection.

8.4 Compliance & Security

- **PCI-DSS** — card data never touches Ayutree servers; Razorpay is the merchant aggregator.
- **Signature** — HMAC-SHA256 protects against tampering.
- **TLS-1.2+** enforced end-to-end.
- **Idempotency** — receipt and signature checks prevent double-charge.
- **GST** — invoice tax line item recorded per order.
- **RBI guidelines** — only Razorpay-supported instruments enabled.

9. Admin Console

The admin console lives at **/admin** behind a role-gated route (role=admin). It uses the same brand palette but with a denser, more functional layout.

9.1 Dashboard Metrics

- **Revenue** — sum of paid orders + COD-shipped/delivered orders.
- **Orders** — total order count.
- **Customers** — total registered customers (role=customer).
- **Products** — total SKUs in catalogue.
- **Sales — Last 7 Days** — daily line chart.
- **Top Products** — top 5 by units sold (bar chart).
- **Low-Stock** — products with stock < 20.

9.2 Product Management

- Search across product names; click to edit.
- Create new product (drawer): name, image URL, category, price, MRP, stock, descriptions, ingredients, featured flag.
- Edit an existing product; changes save in-place.
- Delete with confirmation.

9.3 Order Management

- List all orders (newest first).
- Change status: placed → shipped → delivered (or cancelled).
- Open detail drawer for customer info, line items, payment status, totals.

9.4 Customer Management

- Listing of customers (excludes admins).
- Per-row: order count, lifetime spend, signup method (local / google).

10. Data Model & Database

MongoDB is used as the primary store. Each collection uses a custom UUID-style identifier (**user_id**, **product_id**, **order_id**) rather than MongoDB's ObjectId in the API surface.

users	user_id, email (unique), name, password_hash (bcrypt), role, auth_provider, picture, created_at
products	product_id (unique), name, category, price, mrp, stock, sku, image, short_description, description, ingredients, benefits[], featured, rating, created_at
categories	slug (unique), name, image
carts	user_id (unique), items: [{ product_id, qty }]
wishlists	user_id (unique), product_ids[]
orders	order_id (unique), user_id, user_email, items[], subtotal, shipping, tax, total, address{}, payment_method, payment_status, status, razorpay_order_id?, razorpay_payment_id?, created_at
reviews	review_id, product_id, user_id, user_name, rating (1–5), comment, created_at

Indexes

- users.email — unique
- products.product_id — unique
- orders.order_id — unique

11. API Reference

Auth

- POST /api/auth/register
- POST /api/auth/login
- POST /api/auth/logout
- GET /api/auth/me
- POST /api/auth/google/session

Catalogue

- GET /api/categories
- GET /api/products (?category, ?q, ?featured, ?limit)
- GET /api/products/{product_id}

Cart

- GET /api/cart
- POST /api/cart/add
- POST /api/cart/update
- DELETE /api/cart/{product_id}

Wishlist

- GET /api/wishlist
- POST /api/wishlist/toggle/{product_id}

Orders & Payments

- POST /api/checkout
- POST /api/payments/verify
- GET /api/orders
- GET /api/orders/{order_id}

Reviews

- POST /api/reviews

Admin (role=admin)

- GET /api/admin/stats
- GET /api/admin/orders
- PATCH /api/admin/orders/{order_id}
- GET /api/admin/customers
- POST /api/admin/products

- PATCH /api/admin/products/{product_id}
- DELETE /api/admin/products/{product_id}

12. Security & Compliance

- **Password storage** — bcrypt with per-user salt; never stored as plain text.
- **Session** — JWT (HS256, 7-day TTL), delivered as httpOnly + Secure + SameSite=None cookie, with a Bearer-token fallback for cross-origin contexts.
- **Role gating** — admin routes verify role=admin before any data access.
- **HTTPS-only** — TLS 1.2+ enforced at ingress.
- **CORS** — credentialed origins controlled via env.
- **Input validation** — Pydantic v2 schemas reject malformed bodies.
- **SQL injection** — not applicable (MongoDB document store with parameterised filters).
- **XSS** — React auto-escapes; no *dangerouslySetInnerHTML* in product/admin views.
- **Payment data** — never traverses Ayutree servers (PCI scope: SAQ-A).
- **Webhook signing** — Razorpay signature verified on every callback.
- **Audit** — admin actions logged (planned).
- **Data retention** — Orders & invoices retained ≥ 8 years (GST & Income-Tax Act).

13. Deployment, Scaling & Operations

13.1 Topology

- Frontend SPA served as static assets behind a global CDN.
- Backend FastAPI deployed as a Kubernetes Deployment with HPA (CPU 70%).
- MongoDB as a replica set (Atlas or self-managed) with daily snapshots.
- TLS terminated at ingress; HSTS enabled.

13.2 Environment Variables

MONGO_URL		mongodb://... (connection string)
DB_NAME		ayutree_db
JWT_SECRET		64-char hex
ADMIN_EMAIL	/	Initial admin seed
ADMIN_PASSWORD		
RAZORPAY_KEY_ID	/	from Razorpay Dashboard
KEY_SECRET	/	
WEBHOOK_SECRET		
FRONTEND_URL		https://ayutree.com
REACT_APP_BACKEND_URL		Frontend → backend base URL

13.3 Observability

- Supervisor logs streamed via stdout/stderr.
- Health endpoint: GET /api/categories (200 ⇒ healthy).
- Recommended: Prometheus + Grafana, Sentry, log aggregation (Loki / ELK).

13.4 Backups & DR

- MongoDB daily snapshot + 7-day PITR.
- Object storage versioning enabled; lifecycle policy 30 days.
- Quarterly restore drill.

14. Testing & Quality Assurance

- **Backend** — 25 pytest cases covering catalogue, auth, cart, wishlist, checkout (COD & Razorpay-sim), payment verification, admin CRUD and 403 rejection. *Result: 100% pass.*
- **Frontend** — Playwright e2e: home rendering, catalogue search, PDP add-to-bag, end-to-end customer journey, admin login & CRUD, mobile drawer at 390px. *Result: 100% pass.*
- **Linters** — ESLint (JS) + Ruff (Python) both clean.
- **Manual UAT** — verified across Chrome desktop, Safari iOS, Chrome Android.

Key data-testid attributes (for automation)

- nav-shop-all, nav-cart, cart-count
- product-card-{id}, add-to-cart-{id}, add-to-bag-btn, buy-now-btn
- qty-increment, qty-decrement, qty-value
- login-email, login-password, login-submit, register-*
- place-order-btn, payment-cod, payment-razorpay
- admin-nav-dashboard / products / orders / customers
- edit-{id}, delete-{id}, save-product, order-status-{id}

15. Roadmap & Future Enhancements

P1 (next sprint)

- Object-storage upload for product images in admin.
- Forgot-password flow with secure token email.
- Customer review submission UI on PDP.
- Transactional emails (order placed / shipped / delivered) via Resend.
- WhatsApp order updates (Twilio).

P2 (later)

- Coupon codes & festive promotions.
- Multi-address book + saved payment methods.
- Loyalty / rewards programme.
- Subscriptions & refill auto-orders.
- Hindi & Tamil localisation.
- "Find Your Ritual" ayurvedic dosha quiz with bundle recommendation.
- Influencer affiliate tracking links.
- Shopify-style 'compare at price' & bundle SKUs.

16. Appendix — Credentials & Setup

Default Admin

URL	https://<your-domain>/admin
Email	admin@ayutree.com
Password	Admin@2026 (change immediately in production)

Test Cards (Razorpay sandbox)

Card Number	4111 1111 1111 1111
CVV	123
Expiry	Any future date
OTP	1234 (test environment)
UPI	success@razorpay / failure@razorpay

Local Development

- Backend: `uvicorn server:app --reload --port 8001` (managed by supervisor in this environment).
- Frontend: `yarn start` (port 3000).
- MongoDB: ensure `MONGO_URL` points to a running mongod.
- Seed runs automatically on backend startup (idempotent).

Contact

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This document was generated for the Ayutree e-commerce platform. All technical figures reflect the v1.0 MVP build.